Holiness Trecy Kerandi

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Education

Bucknell University **Expected Jan. 2027**
B.S. in Computer Science & Engineering *Lewisburg, PA*

Awards & Presentations

1st Place, Best-in-Show – Cultural Diffusion Data Challenge Hackathon (2024) **Presenter** – NWSA Conference **Presenter** – Susquehanna River Valley Undergraduate Research Symposium **Upcoming** – "Wild Africa as Data Frontier: AI, Empire, and Decolonial Futures" (Dec. 2026)

Experience

Griot Institute for Black Lives & Cultures, Bucknell University **Sept. 2023 – Present**
Qualitative Research Intern *Lewisburg, PA*

- Structured ∼500 qualitative interviews and archival documents into a tagged Google Sheets dataset using standardized categorization across region, theme, and period, enabling cross-cutting analysis of African girlhood narratives
- Presented at the NWSA Conference and built public-facing Google Sites translating research findings into navigable web experiences for non-academic audiences; sustained 3+ years of interdisciplinary team engagement

International Student Services, Bucknell University **Aug. 2024 & Aug. 2025**
Orientation Guide *Lewisburg, PA*

- Designed CSV-based data workflow system optimizing 90+ onboarding tasks across 50+ countries, reducing manual planning time by 40%
- Built Python scripts reconciling inconsistent staff and student data across multiple spreadsheets, resolving duplicate entries, formatting mismatches, and cross-sheet inconsistencies that previously required manual intervention

Kenya Pipeline Company (KPC) **May 2025 – Aug. 2025**
Software Engineering Intern (Industrial Attaché) *Nairobi, Kenya*

- Developed React and TypeScript frontend components within an enterprise engineering team supporting Kenya's national petroleum infrastructure, contributing to production codebase across software and networking teams

Carceral Algorithms & Data Research, Bucknell University **May 2023 – July 2023**
Undergraduate Research Assistant *Lewisburg, PA*

- Engineered Python/Pandas data analysis pipeline processing 10,000+ misconduct records from the Pennsylvania Department of Corrections, surfacing 3× racial disparity in punishment outcomes across demographic groups
- Built reproducible Jupyter analysis workflows and interactive dashboards visualizing temporal and demographic bias trends; presented findings at the Susquehanna River Valley Undergraduate Research Symposium

Projects

Carceral Data Platform | *React, Python, PostgreSQL, Pandas, ArcGIS JS API, Recharts, React Router* **Aug. 2024 – May 2025**

- Built and shipped full-stack web platform for external client visualizing 50K+ records across 6 monthly datasets at carceral.netlify.app; configured CI/CD pipeline via Netlify enabling automatic production deploys on every merge to main
- Iterated across 10+ client feedback cycles translating non-technical stakeholder requirements into product features, communicating technical tradeoffs across 2 semesters of sustained development
- Developed interactive ArcGIS map with population-scaled facility markers and security-level color encoding; engineered lazy-loading fetch layer requesting only the selected month's CSV on demand for scalable dataset additions

Babe – Privacy-First Wellbeing App | *React Native, Expo, SQLite, JavaScript, TypeScript* **v1 launching 2026**

- Architected privacy-first mobile app with local-only 5-table SQLite schema; deployed via Expo EAS Build with automated OTA updates, targeting a clearly defined underserved user base
- Built custom React Native component library with theme-aware design system (auto light/dark mode); designed shared-state architecture where every mutation triggers SQLite persistence via INSERT-OR-REPLACE upsert logic
- Implemented custom swipe-navigation gestures using PanResponder + Animated API; integrated YouTube oEmbed API for music metadata with graceful text fallback

Key Frenzy – Educational Typing Game | *Java, JavaFX, MVC architecture, Agile/Scrum* **Jan. 2024 – May 2024**

- Built core game-loop logic in a 4-person Scrum team using JavaFX's AnimationTimer and PathTransition; designed for non-English speakers and non-QWERTY keyboard users; authored UML class diagrams modeling MVC separation
- Diagnosed and resolved critical InvalidModule runtime bug blocking game launch; wrote unit test cases for game logic and contributed to code reviews

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, C, HTML/CSS
Frontend & Mobile: React, React Native, JavaFX, responsive design
Backend & Data: PostgreSQL, SQLite, REST API design, Pandas, NumPy, ETL pipelines, normalized schema design, data validation
Cloud & DevOps: Netlify CI/CD, Expo EAS Build, Linux/Unix, Bash, Git
Visualization & Methods: ArcGIS JS API, Recharts, Matplotlib, Tableau, Jupyter, Agile/Scrum, MVC, unit testing